

How to Create and Visualize Metrics in Catalog

Learn how to generate a specific metric view for your catalog and schema using automated tools. This guide covers the process of running queries, configuring permissions, and navigating to the lineage graph to analyze your data effectively.

1 Click this icon.

The screenshot shows the Databricks workspace interface. On the left is a sidebar with navigation options: Home, Workspace (selected), Recents, Catalog, Jobs & Pipelines, Compute, Discover (Beta), Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, and SQL Warehouses. The main area displays a notebook titled '03-metric-view'. The notebook content includes a title 'Metric View: IITB Subreddit Metrics', a description 'This notebook creates a **metric view** for analytics on r/iitbombay subreddit data.', configuration instructions, prerequisites, and two code blocks. The first code block is a Python script for widget setup, and the second is a SQL query to create a metric view.

```
1 # Widget setup - configure your catalog and schema
2 dbutils.widgets.text("catalog", "", "Catalog Name")
3 dbutils.widgets.text("schema", "", "Schema Name")
4
5 catalog = dbutils.widgets.get("catalog")
6 schema = dbutils.widgets.get("schema")
7 print(f"Creating metric view in: {catalog}.{schema}")
```

```
1 %sql
2 CREATE OR REPLACE VIEW ${catalog}.${schema}.iitb_subreddit_metrics
3 WITH METRICS
4 ...
```

2 Click here.

Metric View: IITB Subreddit Metrics

This notebook creates a **metric view** for analytics on r/iitbombay subreddit data.

Configuration: Use the widget inputs at the top of this notebook to set your catalog and schema.

Prerequisites: Run 01-data-ingestion.ipynb and 02-data-transformation/ SQL files first.

```
1 # Widget setup - configure your catalog and schema
2 dbutils.widgets.text("catalog", "", "Catalog")
3 dbutils.widgets.text("schema", "", "Schema")
4
5 catalog = dbutils.widgets.get("catalog")
6 schema = dbutils.widgets.get("schema")
7 print(f"Creating metric view in: {catalog}")
```

```
1 %sql
2 CREATE OR REPLACE VIEW ${catalog}.${schema}.iitb_subreddit_metrics
3 WITH METRICS
4 LANGUAGE YAML
5 AS $$
6 version: 1.1
```

Genie Code

Run multi-step data and AI tasks

What can Genie Code do?

Train a model

Summarize and interpret the notebook

@ for objects, for commands, ↑↓ for hist...

Agent

Only use the agent with code and data you trust

3 Type "create this metric view for catalog "iitb" and schema "bharat_bricks" Enter"

4 Click "Ask every time"

Metric View: IITB Subreddit Metrics

This notebook creates a **metric view** for analytics on r/iitbombay subreddit data.

Configuration: Use the widget inputs at the top of this notebook to set your catalog and schema.

Prerequisites: Run 01-data-ingestion.ipynb and 02-data-transformation/ SQL files first.

```
1 # Widget setup and schema
2 dbutils.widgets.text("catalog", "", "Catalog Name")
3 dbutils.widgets.text("schema", "", "Schema Name")
4
5 catalog = dbutils.widgets.get("catalog")
6 schema = dbutils.widgets.get("schema")
7 print(f"Creating metric view in: {catalog}.{schema}")
```

03-metric-view

create this metric view for catalog "iitb" and schema "bharat_bricks"

Thoughts: The user wants me to create a...

I'll create the metric view for you with catalog "iitb" and schema "bharat_bricks". Let me update the widget configuration and run the cells.

Edited 03-metric-view, updated 1 cell

Thoughts: Good, I've updated the cell to ...

Running 2 cells

Configure catalog and schema

Cell 3

Ask every time Skip Allow

1 asset

03-metric-view +2 -2 X ✓

Configure catalog and sch... +2 -2 X ✓

Send follow-up messages to guide Genie ...

Agent

5 Click "Always allow in current thread"

Metric View: IITB Subreddit Metrics

This notebook creates a **metric view** for analytics on r/iitbombay subreddit data.

Configuration: Use the widget inputs at the top of this notebook to set your catalog and schema.

Prerequisites: Run 01-data-ingestion.ipynb and 02-data-transformation/ SQL files first.

```
1 # Widget setup and schema
2 dbutils.widgets.text("catalog", "", "Catalog Name")
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5 catalog = dbutils.widgets.get("catalog")
6 schema = dbutils.widgets.get("schema")
7 print(f"Creating metric view in: {catalog}.{schema}")
```

03-metric-view

create this metric view for catalog "iitb" and schema "bharat_bricks"

Thoughts: The user wants me to create a...

I'll create the metric view for you with catalog "iitb" and schema "bharat_bricks". Let me update the widget configuration and run the cells.

Edited 03-metric-view, updated 1 cell

Thoughts: Good, I've updated the cell to ...

Running 2 cells

Configure catalog and schema

Cell 3

Ask every time Skip Allow

✓ Ask every time

Always allow in current thread

Always allow

1 asset

03-metric-view +2 -2 X ✓

Configure catalog and sch... +2 -2 X ✓

Send follow-up messages to guide Genie ...

Agent

Always review the accuracy of responses.

6 Click "Run"

The screenshot shows the Databricks workspace interface. On the left is a sidebar with navigation options like Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, Data Ingestion, AI/ML, Playground, AI Gateway, Experiments, Features, and Models. The main area displays a notebook titled 'Metric View: IITB Subreddit Metrics'. At the top, there are input fields for 'Catalog Name' (catalog) and 'Schema Name' (schema). Below these, the notebook content includes a description, configuration instructions, prerequisites, and a code cell. The code cell is highlighted with a blue border, and the 'Run' button is circled in orange. The right sidebar shows a 'Thoughts' section with a message from the AI agent, a 'Running 2 cells' section with a 'Run' button, and a '1 asset' section with a 'Run' button.

Catalog Name
catalog
iitb

Schema Name
schema
bharat_bricks

Metric View: IITB Subreddit Metrics

This notebook creates a **metric view** for analytics on `r/iitb` subreddit data.

Configuration: Use the widget inputs at the top of this notebook to set your catalog and schema.

Prerequisites: Run `01-data-ingestion.ipynb` and `02-data-transformation/ SQL` files first.

```
1 # Widget setup and schema
2 dbutils.widgets.text("catalog", "", "Catalog Name")
3 dbutils.widgets.text("schema", "", "Schema Name")
4
5 catalog = dbutils.widgets.get("catalog")
6 schema = dbutils.widgets.get("schema")
7 print(f"Creating metric view in: {catalog}.{schema}")
```

Running 2 cells
Configure catalog and schema
Cell 3
Review suggested: The code uses `CREATE OR REPLACE VIEW`, which is considered unsafe because it will overwrite an existing view if it exists. Overwriting views is destructive as it removes the previous query definition, which is considered data. No external API calls or permission changes are present, but the destructive nature of `'OR REPLACE'` makes this unsafe.

1 asset
03-metric-view +2 -2
Configure catalog and sch... +2 -2

Send follow-up messages to guide Genie ...
Agent

Always review the accuracy of responses.

7 Click "Accept all"

The screenshot shows the Databricks workspace interface after the 'Run' button was clicked. The code cell now shows the output of the previous run, which was 'Creating metric view in: iitb.bharat_bricks'. The 'Accept all' button is circled in orange. The right sidebar shows a 'Thoughts' section with a message from the AI agent, a 'Running 2 cells' section with a 'Run' button, and a '1 asset' section with a 'Run' button.

Catalog Name
catalog
iitb

Schema Name
schema
bharat_bricks

Metric View: IITB Subreddit Metrics

This notebook creates a **metric view** for analytics on `r/iitb` subreddit data.

Configuration: Use the widget inputs at the top of this notebook to set your catalog and schema.

Prerequisites: Run `01-data-ingestion.ipynb` and `02-data-transformation/ SQL` files first.

```
1 # Widget setup and schema
2 dbutils.widgets.text("catalog", "", "Catalog Name")
3 dbutils.widgets.text("schema", "", "Schema Name")
4
5 catalog = dbutils.widgets.get("catalog")
6 schema = dbutils.widgets.get("schema")
7 print(f"Creating metric view in: {catalog}.{schema}")
```

Creating metric view in: iitb.bharat_bricks

4 steps
Perfect! I've successfully created the metric view `iitb.bharat_bricks.iitb_subreddit_metrics`.
This metric view consolidates `r/iitb` subreddit analytics with:
Dimensions: Post date, month, year, academic term (semester), flair, content type, author, and author affiliation
Measures: Total posts, total comments, average post score, average comments per post, high engagement posts, high engagement rate, and unique authors
The view joins `gold_posts` with aggregated `gold_comments` statistics and is ready to use with AI/BI dashboards, Genie spaces, or SQL queries using the `MEASURE()` syntax.

Query the metric view
Create a dashboard with these metrics
Show sample data

1 asset
03-metric-view +2 -2
Configure catalog and sch... +2 -2

@ for objects, / for commands, ↑↓ for hist...
Agent

Always review the accuracy of responses.

8 Click "Catalog"

The screenshot shows the Databricks workspace interface. On the left sidebar, the 'Catalog' link is highlighted with an orange circle. The main area displays a SQL query editor with the following content:

```
SQL query contains $ parameter. Migrate to the parameter marker syntax, :param:
Parameter values:
• catalog: iitb
• schema: bharat_bricks
```

The right sidebar shows the 'Genie Code' panel with a message: "Perfect! I've successfully created the metric view iitb.bharat_bricks.iitb_subreddit_metrics." It also lists dimensions and measures for the metric view.

9 Click this icon.

The screenshot shows the Databricks workspace interface. On the left sidebar, the 'Catalog' link is highlighted with an orange circle. The main area displays a table of suggested metrics:

Name	Reason for suggestion
gold_comments iitb.bharat_bricks	You view frequently
gold_posts iitb.bharat_bricks	You view frequently
iitb	You view frequently
bharat_bricks iitb	You view frequently
data iitb.bharat_bricks	You view frequently
vishesh	You view frequently
gold_posts_chunked iitb.bharat_bricks	You viewed • 1 hour ago
posts iitb.bharat_bricks	You viewed • 1 hour ago
comments iitb.bharat_bricks	You viewed • 1 hour ago

The right sidebar shows the 'Genie Code' panel with a message: "Perfect! I've successfully created the metric view iitb.bharat_bricks.iitb_subreddit_metrics." It also lists dimensions and measures for the metric view.

10 Click this icon.

The screenshot shows the Databricks Catalog interface. On the left sidebar, the 'Catalog' section is expanded, and the 'bharat_bricks' folder is highlighted with an orange circle. The main panel displays a list of suggested metrics, including 'gold_comments', 'gold_posts', 'iitb', 'bharat_bricks', 'data', 'vishesh', 'gold_posts_chunked', 'posts', and 'comments'. The right sidebar shows the 'Genie Code' section with a code editor and a 'Query the metric view' button.

Name	Reason for suggestion
gold_comments iitb.bharat_bricks	You view frequently
gold_posts iitb.bharat_bricks	You view frequently
iitb iitb.bharat_bricks	You view frequently
bharat_bricks iitb	You view frequently
data iitb.bharat_bricks	You view frequently
vishesh iitb.bharat_bricks	You view frequently
gold_posts_chunked iitb.bharat_bricks	You viewed • 1 hour ago
posts iitb.bharat_bricks	You viewed • 1 hour ago
comments iitb.bharat_bricks	You viewed • 1 hour ago

11 Click this icon.

The screenshot shows the Databricks Catalog interface. On the left sidebar, the 'bharat_bricks' folder is expanded, and the 'Table (8)' icon is highlighted with an orange circle. The main panel displays a list of suggested metrics, including 'gold_comments', 'gold_posts', 'iitb', 'bharat_bricks', 'data', 'vishesh', 'gold_posts_chunked', 'posts', and 'comments'. The right sidebar shows the 'Genie Code' section with a code editor and a 'Query the metric view' button.

Name	Reason for suggestion
gold_comments iitb.bharat_bricks	You view frequently
gold_posts iitb.bharat_bricks	You view frequently
iitb iitb.bharat_bricks	You view frequently
bharat_bricks iitb	You view frequently
data iitb.bharat_bricks	You view frequently
vishesh iitb.bharat_bricks	You view frequently
gold_posts_chunked iitb.bharat_bricks	You viewed • 1 hour ago
posts iitb.bharat_bricks	You viewed • 1 hour ago
comments iitb.bharat_bricks	You viewed • 1 hour ago

12 Click "iitb_subreddit_metrics"

Home

Workspace

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Discover Beta

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Genie

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Runs

Data Ingestion

AJ/ML

Playground

AI Gateway Beta

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Features

Serverless Starter ... 2XS

Type to search...

For you All

My organization

workspace

system

iiitb

bharat_bricks

Tables (8)

comments

gold_comments

gold_posts

gold_posts_chunked

iiitb_subreddit_metrics

posts

silver_comments

silver_posts

Volumes (1)

default

information_schema

vishesh

Delta Shares Received

samples

dbdemos_vishesh

Govern Connect Share Create

Suggested Favorites Recents

Q Filter

Name	Reason for suggestion
gold_comments iiitb.bharat_bricks	You view frequently
gold_posts iiitb.bharat_bricks	You view frequently
iiitb	You view frequently
bharat_bricks iiitb	You view frequently
data iiitb.bharat_bricks	You view frequently
vishesh	You view frequently
gold_posts_chunked iiitb.bharat_bricks	You viewed • 1 hour ago
posts iiitb.bharat_bricks	You viewed • 1 hour ago
comments iiitb.bharat_bricks	You viewed • 1 hour ago
data vishesh.default	You viewed • 4 hours ago
default iiitb	You viewed • 1 hour ago

Perfect! I've successfully created the metric view `iiitb.bharat_bricks.iiitb_subreddit_metrics`.

This metric view consolidates `r/iiitb` subreddit analytics with:

Dimensions: Post date, month, year, academic term (semester), flair, content type, author, and author affiliation

Measures: Total posts, total comments, average post score, average comments per post, high engagement posts, high engagement rate, and unique authors

The view joins `gold_posts` with aggregated `gold_comments` statistics and is ready to use with AI/BI dashboards, Genie spaces, or SQL queries using the `MEASURE()` syntax.

[Query the metric view](#)

[Create a dashboard with these metrics](#)

[Show sample data](#)

1 asset

03-metric-view +2 -2

@ for objects, / for commands, ↑↓ for hist...

Agent

13 Click this button.

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover Beta

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

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Data Engineering

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Type to search...

Overview

Details Permissions Lineage Insights

Description

Unified metric view for the r/iitbombay subreddit — IIT Bombay campus life, academic discussions, and community health dashboards via AI/BI.

Measures (7)

Name	Comment
Total Posts Total Posts	Total number of posts submitted.
Total Comments Total Comments	Sum of comment counts across all posts.
Avg Post Score Avg Post Score	Mean net vote score per post.
Avg Comments per Post Avg Comments per Post	Mean number of comments per post.
High Engagement Posts High Engagement Posts	Posts with 20+ comments or 100+ score.
High Engagement Rate High Engagement Rate	Fraction of posts classified as high-engagement.
Unique Authors Unique Authors	Number of distinct Reddit users who authored posts.

Dimensions (8)

Name	Comment	Tags
Post Date	Date (UTC) the post was	

Total Posts

Total Posts

Total number of posts submitted.

Definition

Type: bigint

Format: Number · 1235

Synonyms: post count, number of posts

Expression

COUNT(1)

Show full details

14 Click "Details"

The screenshot shows the Databricks Catalog Explorer interface. On the left, the 'Catalog' sidebar is open, showing a tree view of the workspace. The 'iitb_subreddit_metrics' table is selected under the 'iitb' workspace. The main panel displays the 'Details' tab for this table. An orange circle highlights the 'Details' link in the top navigation bar. The table description states: 'Unified metric view for the r/iitbombay subreddit — IIT Bombay campus life, academics, placements, hostel culture, politics, and community discussions. Posts are the primary grain, enriched with pre-aggregated comment statistics. Use for campus sentiment, engagement trends, content analysis, and community health dashboards via AI/BI, Genie, or RAG agents.' Below the description, a table of measures is shown, including 'Total Posts', 'Total Comments', 'Avg Post Score', 'Avg Comments per Post', 'High Engagement Posts', 'High Engagement Rate', and 'Unique Authors'. A sidebar on the right provides a definition for 'High Engagement Posts'.

Measures (7)

Name	Comment	Tags
Total Posts Total Posts	Total number of posts submitted.	
Total Comments Total Comments	Sum of comment counts across all posts.	
Avg Post Score Avg Post Score	Mean net vote score per post.	
Avg Comments per Post Avg Comments per Post	Mean number of comments per post.	
High Engagement Posts High Engagement Posts	Posts with 20+ comments or 100+ score.	
High Engagement Rate High Engagement Rate	Fraction of posts classified as high-engagement.	
Unique Authors Unique Authors	Number of distinct Reddit users who authored posts.	

High Engagement Posts
High Engagement Posts
Posts with 20+ comments or 100+ score.

Definition
Type: bigint
Format: Number · 1235
Synonyms: viral posts, trending posts

Expression
SUM(CASE WHEN num_comments >= 20 OR post_score >= 100 THEN 1 ELSE 0 END)

15 Click "Collapse"

The screenshot shows the Databricks Catalog Explorer interface. On the left, the 'Catalog' sidebar is open, showing a tree view of the workspace. The 'iitb_subreddit_metrics' table is selected under the 'iitb' workspace. The main panel displays the 'Details' tab for this table. An orange circle highlights the 'Collapse' button in the bottom right corner of the definition panel. The table definition is shown in YAML format.

Definition YAML

```
version: 1.1

source: |-
  SELECT
    p.post_id,
    p.title,
    p.body,
    p.author,
    p.created_at,
    p.score AS post_score,
    p.upvote_ratio,
    p.num_comments,
    p.flair,
    p.author_flair_text,
    p.is_self,
    p.is_video,
    ... 204 more line
```

16 Click "Lineage"

databricks

New

Home

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover Beta

Marketplace

SQL

SQL Editor

Queries

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iiib

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Tables (8)

comments

gold_comments

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gold_posts_chunked

iiib_subreddit_metrics

posts

silver_comments

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Volumes (1)

default

information_schema

vishesh

Delta Shares Received

samples

dbdemos vishesh

Catalog Explorer > iiib > bharat_bricks >

iiib_subreddit_metrics

Ask Genie Preview Edit Share Create

Overview Details Permissions Lineage Insights

Definition

YAML

version: 1.1

source: |-

SELECT

p.post_id,

p.title,

p.body,

p.author,

p.created_at,

p.score AS post_score,

p.upvote_ratio,

p.num_comments,

p.flair,

p.author_flair_text,

p.is_self,

p.is_video,

p.is_original_content,

p.spoiler,

p.locked,

p.domain,

p.num_crossposts,

p.subreddit_subscribers,

p.content_type,

p.permalink,

COALESCE(c.total_comment_score, 0) AS total_comment_score,

COALESCE(c.avg_comment_score, 0.0) AS avg_comment_score,

COALESCE(c.max_thread_depth, 0) AS max_thread_depth,

17 Click "See lineage graph"

The screenshot displays the Databricks workspace interface. On the left, the 'Catalog' sidebar is visible, showing a tree view of the catalog structure. The 'iitb' database is selected, and the 'iitb_subreddit_metrics' table is highlighted. The main area shows the 'Lineage' view for this table, which is currently empty. A tooltip is visible over the 'See lineage graph' button, stating: 'Use this type selector to switch between entity types to explore additional lineage'. The top navigation bar includes the Databricks logo, a search bar, and workspace information.

18 Click "Hide columns"

The screenshot displays a data catalog interface. On the left is a sidebar with navigation options: Workspace, Recents, Catalog (selected), Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouse, Data Engineering, Runs, Data Ingestion, AI/ML, Playground, AI Gateway, Experiments, Features, and Models. The main workspace shows a diagram of data views. Three nodes are visible: 'iitb.bharat_bricks.gold_comments' (Materialized view), 'iitb.bharat_bricks.gold_posts' (Materialized view), and 'iitb.bharat_bricks.iitb_subreddit_metrics' (Metric view). The 'gold_posts' node is highlighted with an orange circle around its 'Show 13 more columns' link. The 'gold_comments' node shows columns 'comment_id' (string) and 'post_id' (string). The 'gold_posts' node shows columns 'post_id' (string) and 'title' (string). The 'iitb_subreddit_metrics' node shows columns 'Post Date' (date) and 'Post Month' (timestamp). The 'Show 13 more columns' link in the 'gold_posts' node is the target of the instruction.